Type: DES

Q1A. Justify the need for an operating system in the embedded system, showing the benefits of concurrent execution over sequential execution with the help of a program. (5)

Q1B. What is the hex value loaded in the pre-scale register of a timer in LPCA768, with a 1-millisecond resolution and 25MHz PCLK? Use the LPC1768 timer for the delay generation, and write a program to blink the LED continuously. (3)

Q1C. Give the advantages of the Cortex M3 processor over traditional ARM processors. (2)

Q2A. Draw the single blocked queue and multiple blocked queue model diagram. Write the drawbacks of a single queue over multiple queues. How multiple queueing model help in the processes with event-based execution? (5)

Q2B. Write an RTX code for LPC1768 to display the message "Manipal is the dream destination" in one task and blink the four LEDs in another task by using the **event register** function. Discuss the logic used in detail. (5)

Q3A. Draw the timing diagram with comparative analysis for the given five processes with the type of scheduling in **Table 1.** Complete the table with the finish time, Turnaround Time (TAT), mean TAT and normalised TAT. (5)

Table.1

Process	**A**	**B**	**C**	**D**	**E**	**Mean**
Arrival time	0	2	3	5	7	
Service time (Ts)	3	2	4	4	3	
SRT						
Finish Time						Mean
TAT (Tt)						
Tt/ Ts						
HRRN						
Finish Time						Mean
TAT (Tt)						
Tt/ Ts						

(5)

Q3B. Discuss the problem due to simple Feedback scheduling and how to overcome that. Draw the timing diagram to analyze the feedback scheduling for the five processes given in **Table 2.** Complete the table with the finish time, Turnaround Time (TAT) for all the processes, and mean TAT and normalised TAT.

Table 2

Process	**A**	**B**	**C**	**D**	**E**	**Mean**
Arrival time	0	2	3	5	7	
Service time (Ts)	3	2	4	4	3	
FB q=1						
Finish Time						Mean
TAT (Tt)						
Tt/ Ts						
FB q=2^i						
Finish Time						Mean
TAT (Tt)						
Tt/ Ts						

Q4A. The two processes P and Q have an execution sequence of get A, get B, Release A, and release B for process P and get B, get A, Release A, and release B for process Q . Using a diagram, explain the possible situations of deadlock and also write the instruction sequence that can avoid the deadlock situation. (5)

Q4B. Demonstrate how the MUTEX function is used in RTX code for LPC1768 to print the value of a global variable updated in each task. Assume there are 2 tasks. Write comments to explain the logic. (5)

Q5A. Discuss the any four task-switching mechanisms in the pre-emptive scheduling. (4)

Q5B. What is dynamic memory allocation and write how it is used in the Free RTOS? (3)

Q5C. Discuss message-passing mechanisms and mailbox functions in RTX. (3)